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GitHub Link	https://github.com/mani759/deep-learning

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

**COURSE NAME: Deep Learning COURSE CODE: MLA0405 Faculty : Dr. Arul Raj
A.M Slot : D**

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

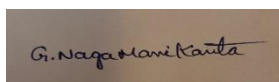
Duration: 90 minutes

Marks: 25

Code of Conduct:

I _____ G.Naga Manikanta(192425310)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: G.Naga Manikanta

Scenario-Based Questions CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer

For identifying abnormal ECG patterns from thousands of **unlabeled ECG records**, the most suitable deep learning technique is an **Autoencoder**, which is an unsupervised neural network.

Why Autoencoders?

In this scenario, the hospital does not have labeled data indicating whether an ECG is normal or abnormal. Since autoencoders learn patterns without requiring labels, they are ideal for this task. They are trained to reconstruct input data by learning the important features of normal ECG signals. When an abnormal ECG is given as input, the model cannot reconstruct it accurately, resulting in a high reconstruction error. This error is then used to identify abnormal heart patterns.

Workflow

1. **Collect ECG Data:** Gather thousands of unlabeled ECG recordings from patients.
2. **Preprocess the Data:** Remove noise, normalize the signals, and divide them into fixed-length segments.
3. **Train the Autoencoder:** Feed the ECG signals into the autoencoder. The encoder compresses the input into a smaller feature representation, while the decoder reconstructs the original ECG signal.
4. **Calculate Reconstruction Error:** Compare the original ECG with the reconstructed ECG using an error metric such as Mean Squared Error (MSE).
5. **Detect Anomalies:** If the reconstruction error is greater than a predefined threshold, the ECG is classified as abnormal; otherwise, it is considered normal.

6. **Medical Review:** Cardiologists examine the detected abnormal ECGs for diagnosis and treatment.

Advantages

- Works without labeled training data.
- Learns meaningful features automatically.
- Detects rare and unusual ECG patterns effectively.
- Reduces the workload of healthcare professionals by automatically filtering suspicious cases.

Limitations

- Performance depends on the quality and diversity of the training data.
- Choosing the correct anomaly threshold is important.
- Some rare normal ECG variations may be incorrectly detected as abnormal.

Conclusion

Autoencoders are an effective unsupervised deep learning technique for ECG anomaly detection because they learn the normal characteristics of ECG signals without requiring labeled data. By measuring reconstruction error, they can accurately identify abnormal heart patterns, making them valuable in large-scale hospital monitoring systems.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer

A retail company can use an **Autoencoder** to reduce the dimensionality of customer purchase data before performing customer segmentation. An autoencoder is an **unsupervised neural network** that learns a compressed representation of the input data while preserving its important features.

Why Use an Autoencoder?

Customer purchase data often contains many features such as product categories, purchase frequency, spending amount, discounts, and shopping preferences. High-dimensional data increases computational cost and may contain redundant information.

An autoencoder compresses this data into a smaller set of meaningful features, making customer segmentation faster and more accurate.

Workflow

1. **Collect Customer Data:** Gather customer purchase records, including transaction history, spending habits, and product preferences.
2. **Preprocess the Data:** Clean missing values, normalize numerical features, and encode categorical variables if required.
3. **Train the Autoencoder:** Feed the customer data into the autoencoder. The encoder compresses the input into a lower-dimensional representation (latent features), while the decoder reconstructs the original data.
4. **Extract Latent Features:** Use the compressed representation generated by the encoder as the new feature set.
5. **Perform Customer Segmentation:** Apply clustering algorithms such as **K-Means** or **Hierarchical Clustering** on the compressed data to group customers with similar purchasing behavior.
6. **Analyze Customer Groups:** Use the identified customer segments for targeted marketing, personalized recommendations, and business decision-making.

Advantages

- Reduces the number of features while preserving important information.
- Removes redundant and noisy data.
- Improves the efficiency of clustering algorithms.
- Helps discover hidden patterns in customer purchasing behavior.

Limitations

- Requires sufficient training data for effective feature learning.
- The compressed representation may lose some information if the bottleneck is too small.
- Training can be computationally expensive for very large datasets.

Conclusion

Autoencoders provide an efficient method for reducing the dimensionality of customer purchase data by learning compact feature representations. The compressed data can then be used for customer segmentation, enabling the retail company to identify customer groups more accurately and improve marketing strategies.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer

A facial recognition system that performs poorly with a **shallow neural network** can be improved by increasing the **depth of the network** and using suitable **activation functions**. Deep neural networks are capable of learning complex patterns and facial features more effectively than shallow networks.

Why Increase Network Depth?

A shallow neural network has only one or two hidden layers and may not capture the complex features required for facial recognition. By adding more hidden layers, the network can learn low-level features (such as edges and textures), mid-level features (such as eyes, nose, and mouth), and high-level features (such as complete facial structures). This improves recognition accuracy.

Recommended Architectural Changes

1. **Increase the Number of Hidden Layers:** Use a deeper neural network with multiple hidden layers to learn complex facial features.
2. **Increase the Number of Neurons:** Add more neurons in each hidden layer to improve the model's learning capacity.
3. **Use ReLU Activation Function:** Apply the Rectified Linear Unit (ReLU) in hidden layers because it speeds up training and reduces the vanishing gradient problem.
4. **Use Softmax Activation in the Output Layer:** If the system identifies one person from multiple known individuals, use the Softmax activation function to classify the input into the correct class.
5. **Train and Evaluate the Model:** Train the deeper network using facial image datasets, validate its performance, and fine-tune the architecture if required.

Advantages

- Learns complex facial patterns more effectively.
- Improves recognition accuracy.
- ReLU speeds up training and supports deep networks.
- Better generalization for real-world facial images.

Limitations

- Requires more training data.
- Increases computational cost and training time.
- May overfit if regularization techniques are not used.

Conclusion

Replacing a shallow neural network with a deeper architecture and using **ReLU** activation in the hidden layers and **Softmax** in the output layer significantly improves facial recognition performance. The deeper network learns hierarchical facial features, resulting in higher accuracy and better real-world performance.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer

A **Restricted Boltzmann Machine (RBM)** is an **unsupervised neural network** that can be used to build a movie recommendation system without requiring labeled data. It learns hidden patterns in users' viewing history and predicts movies that users are likely to watch or enjoy.

Why Use an RBM?

A streaming service stores information about the movies each user has watched or rated. Since there are no labels indicating which movie should be recommended, an RBM can learn the relationships between users and movies automatically. It discovers hidden preferences, such as favorite genres, actors, or viewing habits, and uses them to generate personalized recommendations.

Workflow

1. **Collect User Data:** Gather users' viewing history, ratings, or watch lists.
2. **Create User-Movie Matrix:** Represent the data as a matrix where rows represent users and columns represent movies.
3. **Train the RBM:** The visible layer contains user viewing or rating information, while the hidden layer learns latent features such as movie genres and user interests.

4. **Learn Hidden Patterns:** The RBM identifies relationships between users and movies by learning common viewing behaviors.
5. **Predict Missing Preferences:** Estimate the likelihood that a user will enjoy movies they have not yet watched.
6. **Generate Recommendations:** Recommend the movies with the highest predicted preference scores to each user.

Advantages

- Does not require labeled training data.
- Learns hidden user preferences automatically.
- Generates personalized movie recommendations.
- Handles large user and movie datasets efficiently.

Limitations

- Requires significant computational resources for large datasets.
- Training can be slow with millions of users and movies.
- Performance depends on the quality of viewing history data.

Conclusion

Restricted Boltzmann Machines are an effective unsupervised learning technique for movie recommendation systems. By learning hidden patterns from users' viewing history, RBMs can accurately predict user preferences and provide personalized movie recommendations without requiring labeled data.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer

A manufacturing company can use a **deep neural network with representation learning** to detect defective products from sensor readings collected every second. Representation learning enables the model to automatically learn important features from raw sensor data instead of relying on manually designed features, improving the accuracy of defect detection.

Why Use Representation Learning?

Sensor data generated during manufacturing is often high-dimensional and complex. Traditional methods require manual feature extraction, which may miss important patterns. A deep neural network automatically learns meaningful representations of the sensor data, making it easier to distinguish between normal and defective products.

Workflow

1. **Collect Sensor Data:** Gather sensor readings such as temperature, pressure, vibration, and speed from the manufacturing process.
2. **Preprocess the Data:** Remove noise, normalize the values, and organize the readings into fixed time intervals.
3. **Train the Neural Network:** Feed the processed sensor data into a deep neural network. The hidden layers automatically learn useful features and patterns through representation learning.
4. **Classify Products:** The output layer classifies each product as **Normal** or **Defective** based on the learned features.
5. **Evaluate the Model:** Measure performance using metrics such as accuracy, precision, recall, and F1-score.
6. **Deploy the Model:** Use the trained model to monitor sensor readings in real time and automatically identify defective products during production.

Advantages

- Automatically learns important features from raw sensor data.
- Detects defects with high accuracy.
- Supports real-time monitoring of manufacturing processes.
- Reduces manual inspection and improves product quality.

Limitations

- Requires a large amount of training data.
- Training deep neural networks requires high computational resources.
- Performance may decrease if sensor data is noisy or incomplete.

Conclusion

A deep neural network using **representation learning** is an effective solution for detecting defective products from sensor data. By automatically learning meaningful patterns and classifying products as normal or defective, the system improves quality control, reduces manufacturing errors, and increases production efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand